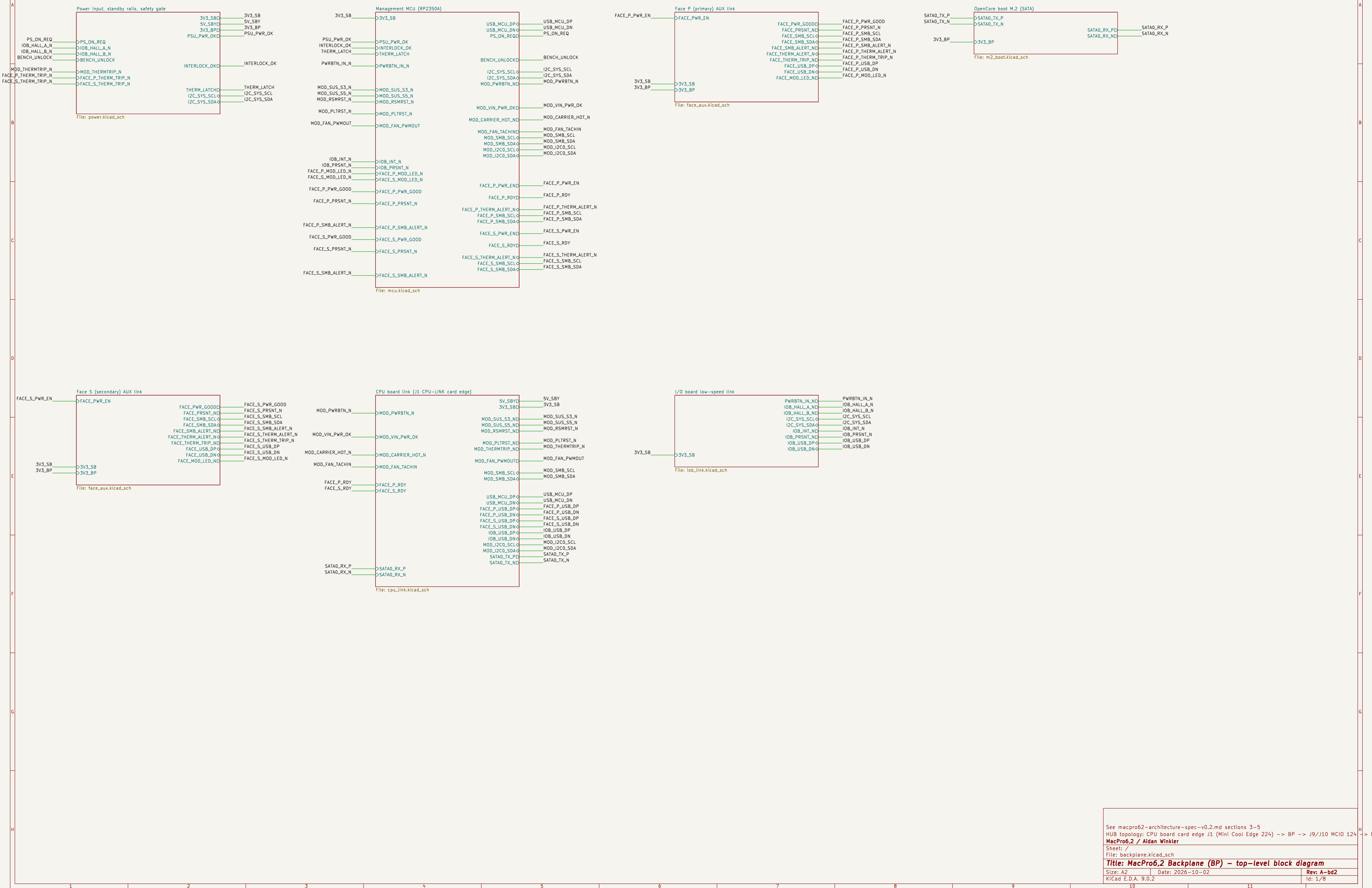


MP62 BACKPLANE rev A – BLOCK DIAGRAM. Sheets are stubs (labels only). Nets with the same label name are connected.

Off-board links: J1 CPU-LINK (card edge from CPU board) | J2 PSU-IN | J3/J4 Face P/S AUX (GH15) | J6 IOB-LINK (fan via IOB CONN_C, no J5) | J7 M.2 OpenCore | J8 SWD | J9/J10 MCIO 124 to Face P/S (PCIe hub)



	1
3V3_SB	1
5V_SBY	2
3V3_BP	3
PSU_PWR_OK	4
PS_ON_REQ	5
IOB_HALL_A_ND	6
IOB_HALL_B_ND	7
BENCH_UNLOCK	8
INTERLOCK_OK	9
MOD_THERMTRIP_ND	10
FACE_P_THERM_TRIP_ND	11
FACE_S_THERM_TRIP_ND	12
THERM_LATCH	13
I2C_SYS_SCL	14
I2C_SYS_SDA	15

BLOCK STUB (rev A block diagram). Contents planned:

- J2 PSU-IN Micro-Fit 3.0 2x4; 12V x2, GND x3, 11V_SB, PS_ON#, PWR_OK (stock PSU pinout TBD, adapter harness)
- 11V_SB -> 3V3_SB (MCU, EEPROMs, face aux in S5); 11V_SB -> 5V_SBY (COM-HPC suspend well)
- 12V_MAIN -> 3V3_BP buck (S0); input TVS + fuse; INA-class monitor on I2C_SYS (12V_MAIN / 11V_SB stay inside this sheet)
- BP 12 V load ~1 A (3V3_BP incl. Gen5 redrivers); fan moved to IOB CONN_C, CB / faces / IOB 12 V by their own PSU leads
- HW gate: PS_ON = INTERLOCK_OK AND PS_ON_REQ AND NOT THERM_LATCH (discrete logic, 3V3_SB)
- THERM_LATCH set by MOD_THERMTRIP_N / FACE_x_THERM_TRIP_N; cleared by AC cycle (TBD)
- INTERLOCK_OK = HALL_A AND HALL_B (or BENCH jumper AND MCU unlock)

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /Power input, standby rails, safety gate/
File: power.kicad_sch

Title: MP62 backplane: Power input, standby rails, safety gate

Size: A4	Date: 2026-10-02	Rev: A-bd2
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[illegible]





	1	2	3	4	5	6
A	5V_SBY<4 3V3_SB<2 MOD_PWRBTN_ND<3 MOD_SUS_S3_N<4 MOD_SUS_S5_N<5 MOD_RSMRST_N<6 MOD_VIN_PWR_OK<7 MOD_PLTRST_N<8 MOD_THERMTRIP_N<9 MOD_CARRIER_HOT_ND<10 MOD_FAN_PWMOUT<11 MOD_FAN_TACHIND<12 MOD_SMB_SCL<13 MOD_SMB_SDA<14 FACE_P_RDY<15 FACE_S_RDY<16 USB_MCU_DP<17 USB_MCU_DN<18 FACE_P_USB_DP<19 FACE_P_USB_DN<20 FACE_S_USB_DP<21 FACE_S_USB_DN<22 IOB_USB_DP<23 IOB_USB_DN<24 MOD_I2C0_SCL<25 MOD_I2C0_SDA<26 SATA0_TX_P<27 SATA0_TX_N<28 SATA0_RX_P<29 SATA0_RX_ND<30	5V_SBY 3V3_SB MOD_PWRBTN_N MOD_SUS_S3_N MOD_SUS_S5_N MOD_RSMRST_N MOD_VIN_PWR_OK MOD_PLTRST_N MOD_THERMTRIP_N MOD_CARRIER_HOT_N MOD_FAN_PWMOUT MOD_FAN_TACHIN MOD_SMB_SCL MOD_SMB_SDA FACE_P_RDY FACE_S_RDY USB_MCU_DP USB_MCU_DN FACE_P_USB_DP FACE_P_USB_DN FACE_S_USB_DP FACE_S_USB_DN IOB_USB_DP IOB_USB_DN MOD_I2C0_SCL MOD_I2C0_SDA SATA0_TX_P SATA0_TX_N SATA0_RX_P SATA0_RX_N	BLOCK STUB (rev A block diagram). Contents planned: – J1: Amphenol Mini Cool Edge 224 (ME1022410103011) vertical SMT; MP62 hub pinout (docs/cpulink_224_pinout_draft.csv, CB mapping column) – PCIe HUB: J1 x16 -> J9 MCIO 124 (Face P), J1 x4 -> J10 MCIO 124 (Face S), REFCLK P/S; 85 ohm L1 (TX) / L6 (RX) – Optional Gen5 build: U10–U13 (Face P) + U14 (Face S) TI DS320PR810 linear redrivers + 220 nF output AC caps – Gen4 build: no redrivers, no AC caps on BP (host–TX caps on module/CPU board, device–TX caps on face) – Also carries: power/state sideband (COM–HPC names, CB = Z790 PCH + EC), fan PWMOUT/TACHIN (demand only), SMBus, I2C0, USB2 x4, SATA0, 5V_SBY x6, 3V3_SB x2, GND x80 – USB2 x4 = MCU (BP MCU), FACEP / FACES (face AUX pins 13/14), SPARE -> IOB–LINK J6 13/14 (IOB hub H2: USB–A A1–A4 + Bluetooth) – BP makes PERST#_x = PLTRST# AND FACE_x_RDY (MCU asserts after FACE_x_PWR_GOOD); CLKREQ#_x end on BP; WAKE#_x ORed to WAKE0# – 12 V main for the CPU board does NOT pass through J1 (CB LUG1/LUG2 single 12 V entry, U11 eFuse)			
B						
C						
D						

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n		
Sheet: /CPU board link (J1 CPU-LINK card edge)/ File: cpu_link.kicad_sch		
Title: MP62 backplane: CPU board link (J1 CPU-LINK card edge)		
Size: A4	Date: 2026-10-02	Rev: A-bd2
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PWRBTN_IN_N	1	PWRBTN_IN_N
IOB_HALL_A_N	2	IOB_HALL_A_N
IOB_HALL_B_N	3	IOB_HALL_B_N
I2C_SYS_SCL	4	I2C_SYS_SCL
I2C_SYS_SDA	5	I2C_SYS_SDA
IOB_INT_N	6	IOB_INT_N
IOB_PRSN_T_N	7	IOB_PRSN_T_N
IOB_USB_DP	8	IOB_USB_DP
IOB_USB_DN	9	IOB_USB_DN
3V3_SBD	10	3V3_SB

BLOCK STUB (rev A block diagram). Contents planned:

- J6 JST GH 15P (BM15B vertical) <-> IOB J6, 1:1 GH 15P cable (ICD 2026-10-02, pinout = IOB plan 5.3):
- 1,2 3V3_SB (BP -> IOB) | 3,7,12,15 GND | 4 PWRBTN_IN_N | 5 HALL_A_N | 6 HALL_B_N | 8 I2C_SYS_SCL | 9 I2C_SYS_SDA
- 10 IOB_INT_N (OD, IOB -> BP, BP pull-up) | 11 IOB_PRSENT_N (IOB ties to GND; BP 10k pull-up to 3V3_SB) | 13/14 USB2_LINK D+/D- (CPU-LINK USB2_SPARE)
- IOB high-speed (USB3 x10, DDI, 2x i226 PCIe, VBAT) runs CB J3 -> IOB HS1 directly; 12 V via the IOB 12P PSU header, not the BP

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /I/O board low-speed link/
File: iob_link.kicad_sch

SATA0_TX_P	1	SATA0_TX_P
SATA0_TX_N	2	SATA0_TX_N
SATA0_RX_P	3	SATA0_RX_P
SATA0_RX_N	4	SATA0_RX_N
3V3_BP	5	3V3_BP

BLOCK STUB (rev A block diagram). Contents planned:

- J7 M.2 M-key socket wired SATA-only (accepts B+M SATA 2242/2230 cards); dedicated OpenCore device
- Set as the default UEFI boot entry (LauncherOption=Full); removable-path fallback on the same ESP
- SATA0 from the CB Z790 PCH via J1 (SATA-capable HSI0 not shared with the M.2 RP9-12 lanes, TBD on the CB)

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /OpenCore boot M.2 (SATA)/
File: m2_boot.kicad_sch

Title: MP62 backplane: OpenCore boot M.2 (SATA)

Size: A4	Date: 2026-10-02
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Rev: A-bd2

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